

Data-based matched-mode source localization for a moving source

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A data-based matched-mode source localization method is proposed in this paper for a moving source, using mode wavenumbers and depth functions estimated directly from the data, without requiring any environmental acoustic information and assuming any propagation model. The method is in theory free of the environmental mismatch problem because the mode replicas are estimated from the same data used to localize the source. Besides the estimation error due to the approximations made in deriving the data-based algorithms, the method has some inherent drawbacks: (1) It uses a smaller number of modes than theoretically possible because some modes are not resolved in the measurements, and (2) the depth search is limited to the depth covered by the receivers. Using simulated data, it is found that the performance degradation due to the afore-mentioned approximation/limitation is marginal compared with the original matched-mode source localization method. The proposed method has a potential to estimate the source range and depth for real data and be free of the environmental mismatch problem, noting that certain aspects of the (estimation) algorithms have previously been tested against data. The key issues are discussed in this paper.

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I. INTRODUCTION

Matched field processing (MFP) was introduced sometimes ago as a method to localize the source in range and depth using acoustic data received on a vertical line array (VLA).¹ The received data are steered by the calculated field (for an assumed source position) using a propagation model, referred to as the replica field, given known sound speed and bottom profiles. Source location is determined by the peak of the range-depth ambiguity function where the calculated (replica) field best matches the data field.^{2,3} When the sound propagation is accurately modeled, as in a slowly varying environment, matched field processing yields good source localization results.⁴ In a temporally varying, and range dependent environment, the results are not as good (i.e., in error) because the sound speed and bottom profiles are not exactly known as a function of range and time. Often there appear many high level sidelobes making the identification of source location ambiguous. This is known as the environmental mismatch problem—due to sparse and not real time environmental measurements. One solution is to estimate the environmental parameters and source parameters simultaneously based on the received data.⁵ The number of parameters that can be searched for is restricted by the independent degrees of freedom in the input data, limiting the search usually to a range independent environment. Others have proposed modified (signal processing) methods designed to tolerate environmental mismatches. They include: (1) Using multiple neighboring location constraints⁶ or sound-speed perturbation constraints⁷ to improve the robustness of

the matched field solution and (2) assigning *a priori* probability distributions of the unknown source location and propagation parameters and then approximating the maximum *a posteriori* probability distribution of the source location, given the received data.⁸ These methods generally involve intensive computations and address, to a large extent, the performance related to the main lobe (level and position uncertainty). In practice, the problem of incorrect source localization often comes from anomalous peaks (at locations far away from the true source range and depth) due to environmental mismatch, noise, and/or interference, etc., resulting in fluctuating and inconsistent answers over the measurement time. In other words, the robustness of the estimations is not assured. This aspect of the problem (the mismatch) may require an information theoretical approach.⁹

Source localization using normal mode theory has drawn much attention in shallow water applications due to the modal nature of sound propagation, particularly at low frequencies. Shang *et al.* pointed out that the source range can be estimated using the phase difference between the modes.¹⁰ Similarly, source depth can be estimated from the ratio of excitation of different modes or from the closure relation of normal modes given that source range has been estimated.¹¹ Yang applied modal beamforming to the mode amplitudes, which were estimated by applying modal decomposition to the VLA data,¹² where to avoid mode estimation bias, he used eigenvector decomposition for the covariance matrix of the mode depth functions. This method can also be considered as matched mode processing (MMP) because the mode steering vector is the calculated mode amplitudes of the replica field. The main difference (between this and earlier work) is that MMP produces an array gain¹³ (similar to MFP) to enhance the signal-to-noise

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ratio (SNR), providing better estimates in low SNR cases. Source localization was demonstrated with Arctic data taking advantage of its stable sound speed profile. It was the first published demonstration of matched mode/field processing. One advantage of MMP is the mode filter, i.e., the ability to drop (filter) the modes that are sensitive to environmental mismatches at the expense of some (slight) degradation in the array gain. (For example, in the Arctic case, one drops the high order modes that interact with the ice cover.) To assure the quality of mode decomposition and increase the array gain, a relatively large aperture VLA is usually used.

For a review of the variety of MFP work before 1993, one is referred to Refs. 2 and 3 for comprehensive discussions.

The vulnerability of MFP/MMP to environmental mismatches has suggested alternative approaches to source location that are model independent. A well-known model-independent approach is time reversal, which focuses the signal to the source position based on the probe signal emitted by the source.¹⁴ For passive detection, one uses a guide source (or a source of opportunity) as the probe source.^{15–17} In this case, one passive-conjugates the receiver data with the signal from the guide source to create “data” at a virtual receiver. The environmental dependence is reduced (largely removed) if the guide source is located at short distances to the target. In practice, the guide source is not always available and is often located at a different depth than the target.

In this paper, we present a model-independent approach to source localization and tracking, assuming a moving source. It is conducted in the mode domain using mode wavenumbers and depth functions directly estimated from the data without requiring any environmental information and assuming any propagation model. In the phone space, it is a model-independent MFP method. As the name implies, the main advantage of this method is that it is potentially free of the environmental mismatch problem and is applicable to (slowly) time-varying environments if the mode parameters (wavenumbers and depth functions) are estimated from the data emitted from the target of interest—the target to be localized and tracked. One can also use data of opportunity to estimate the modal parameters if the data come from a source in the proximity of the target (in range, depth, or time). The latter case will not be studied here.

The method can also be applied to a reduced-aperture or sparsely populated VLA as is the case for MFP with a degraded performance (lower signal or array gain) with the following limitations: Only the source the depth of which is within the depth span of the receiving VLA can be localized, and the depth resolution is limited by the phone spacing, or the Cramer-Rao bound,¹⁸ whichever is larger. In contrast, the conventional MFP does not have the additional constraints.

The paper is organized as follows. In Sec. II, one briefly reviews the Hankel transform that was used to estimate the mode wavenumbers by integrating the narrowband signal (from a moving source) over range in a range independent or adiabatic environment.^{19,20} The method is generalized in Sec. III to a VLA, without *knowing* the source range, to

estimate the mode wavenumber and, in addition, the depth functions. The estimated mode wavenumbers and depth functions are then used to estimate the source location by matched mode/field processing. One notes that the data-based generalized “Hankel” transform is an approximation of the original Hankel transform; as a result, the estimated wavenumbers and depth functions may not be accurate compared with the true values. The question is how the approximation affects the source localization performance compared with the original matched mode/field processing with no environmental mismatch. This issue is investigated using numerical simulations in a summer sound speed profile (SSP) assuming long VLAs with different phone spacing. As is well known, source localization with a narrowband signal often suffers from the sidelobe problem, and the usual practice (to overcome this problem) is to use a broadband signal. With this in mind, we propose a data-based broadband source localization method in Sec. IV. For coherent broadband signals (with a well-defined waveform), one can repeat the preceding processing for each frequency components, and sum the range and depth ambiguity functions. For practical applications, the more challenging and interesting problem is to localize a target using the broadband noise radiated by the target, which is random and incoherent in nature, and for which the Hankel transform does not work. Assuming that the target also radiates narrowband signals, from which one estimates the mode wavenumbers and depth functions at the given frequencies, we propose a data-based method for source localization using the incoherent broadband frequencies. To estimate the mode wavenumbers for the broadband frequency components, we use an interpolation formula based on the equation for the mode group velocity. To interpolate the mode depth functions between frequencies, we employ the WKB approximation for the mode depth functions. The approximations are discussed, and the performance degradations are evaluated for the broadband case using simulation data. In Sec. VI, one studies the localization performance for a short VLA and its limitations. A short summary for the data-based approach is given in Sec. VI. This approach has a potential to be applied to real data. Signal processing issues for real data are discussed in this section.

II. MODE WAVENUMBER ESTIMATION USING HANKEL TRANSFORM

The Hankel transform provides a general theoretical framework for studying acoustic propagation due to a narrowband point source in a horizontally stratified medium. The solution of the Helmholtz equation can be expressed as the zero-order Hankel transform of the depth-dependent Green’s function, $g(k_r)$, given by $p(r) = \int_0^\infty g(k_r)J_0(k_r r)k_r dk_r$ and vice versa, $g(k_r) = \int_0^\infty p(r)J_0(k_r r)rdr$, where J_0 is the Bessel function of order zero.^{19,20} At large ranges, $k_r r \gg 1$, the Hankel transform can be approximated by a Fourier transform, given as follows:^{19,20}

$$p(r) \sim \frac{e^{-i\pi/4}}{\sqrt{2\pi r}} \int_0^\infty g(k_r) e^{-k_r r} \sqrt{k_r} dk_r, \quad k_r r \gg 1, \quad r > 0, \quad (1)$$

$$g(k_r) \sim \frac{e^{i\pi/4}}{\sqrt{2\pi k_r}} \int_0^\infty p(r) e^{ik_r r} \sqrt{r} dr, \quad k_r r \gg 1, \quad k_r > 0. \quad (2)$$

Using the normal mode representation for the pressure field,

$$p(r, z) = \sum_{m=1}^M A_m(r) \phi_m(z),$$

$$A_m(r) = \sqrt{\frac{2\pi}{k_m r}} \phi_m(z_s) e^{-ik_m r - \alpha_m r - i\pi/4}, \quad (3)$$

where k_m and α_m are the mode wavenumber and attenuation coefficient, ϕ_m is the mode depth function, A_m denotes the mode amplitude of the m th mode, and z_s and z denote the source and receiver depth, respectively, one obtains for a finite range R ,

$$g(k_r) \sim \sum_{m=1}^M \frac{\phi_m(z_s) \phi_m(z)}{\sqrt{k_r k_m}} \int_0^R e^{i(k_r - k_m - i\alpha_m)r} dr$$

$$\simeq \sum_{m=1}^M a_m \frac{\phi_m(z_s) \phi_m(z)}{k_r - k_m - i\alpha_m}, \quad (4)$$

where

$$a_m = \frac{e^{i(k_r - k_m)R - \alpha_m R} - 1}{ik_m}. \quad (5)$$

It was observed that the modal wavenumber spectrum, given by Eq. (4), contains many peaks, which occur when the wavenumber equals the mode wavenumbers, noting that the mode attenuation is generally orders of magnitude smaller than the mode wavenumbers. Thus one can measure/estimate the mode wavenumbers from the peaks of the wavenumber spectrum by taking Fourier (wavenumber) transform of the pressure field with respect to range.^{19,20} The method has been generalized to weakly range dependent environments where adiabatic normal modes are valid and tested against experimental data in various ocean environments.^{19–21}

III. DATA-BASED MODAL BEAMFORMING

To illustrate the data-based source localization method, we shall consider a range independent environment and assume that the source moves at a constant speed during the observation time. The source range is unknown, and its depth can be slowly changing with time; for the present discussion, we shall assume it is at a fixed depth. The source geometry relative to the VLA is shown in Fig. 1. We assume that the source radiates narrowband tones (for narrowband source localization) and broadband noise-like signals (for broad band source localization). By Fourier analysis (using long time integration), one obtains a high resolution estimate of the Doppler shifts of the narrowband signals, from which one determines: (1) The *radial* speed of the target, $v = C(\Delta f/f)$, where C is the average sound speed, Δf is the Doppler shift, and f is the known source frequency; and (2) the *radial* range traveled (the distance traveled projected

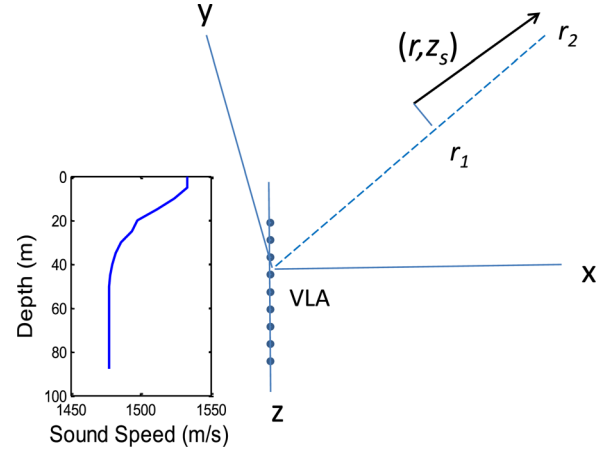


FIG. 1. (Color online) The source track (solid line) relative to the VLA with an inset of the sound speed profile.

along the radial direction) over a time period T , $\Delta R = vT$. The data received by the VLA over time T are used to estimate the mode wavenumber and depth functions, which are assumed time-invariant during this period. To estimate the source range and depth, the data are divided into K blocks. For each block of data, one applies the data-based source localization algorithm to estimate the range and depth using the narrowband and/or broadband signal components. The results can be used to track the source range and depth as a function of time. The key issues are the accuracies of the estimated range and depth, and the (degree of) ambiguities of the estimation; for the latter one investigates the sidelobe-to-main lobe ratio of the ambiguity function.

A. Estimation of the mode wavenumbers and mode depth functions

Because the source range is unknown, one cannot use the wavenumber transform, Eq. (2), to estimate the mode wavenumbers. The modified equation to use is given as follows,

$$\bar{g}(k_r, z_j) = \frac{e^{i\pi/4}}{\sqrt{2\pi k_r}} \int_{r_1}^{r_2} p(r, z_j) e^{ik_r r} S(r) dr, \quad k_r r_1 \gg 1, \quad (6)$$

where $\bar{g}$ denotes the estimated Green's function, $p(r, z_j)$ is a narrowband signal received on the j th receiver, $r = r_1 + vt$, where $t = t_i$, for the i th block of data, r_1 is the unknown initial source range, and $r_2 = r_1 + \Delta R$. The function $S(r)$ is intended to compensate for the spreading loss and will be determined from data using

$$S(r) = \left[\sum_{j=1}^N |p(r, z_j)|^2 \Delta z \right]^{-1/2}, \quad (7)$$

where Δz is the phone spacing. Given a large aperture VLA, one finds based on Eq. (3) and the mode orthogonality relation

$$\sum_{j=1}^N \phi_m(z_j) \phi_l(z_j) \Delta z = \delta_{ml}, \quad (8)$$

that

$$S(r) = \sqrt{\frac{r}{2\pi}} \left[\sum_{m=1}^M \phi_m^2(z_s) \frac{e^{-2\alpha_m r}}{k_m} \right]^{-1/2} = \sqrt{\frac{r}{2\pi}} \tilde{S}(r), \quad (9)$$

where, one notes that $\tilde{S}(r) = e^{\alpha_1 r_1} [\sum_{m=1}^M \phi_m^2(z_s) (e^{-2(\alpha_m - \alpha_1)r_1} \times e^{-2\alpha_m(r-r_1)}/k_m)]^{-1/2}$ varies slowly with respect to r and can be approximated by a constant because $0 \leq (\alpha_m - \alpha_1)r_1 \ll 1$ and $0 \leq \alpha_m(r - r_1) \leq \alpha_m \Delta R \ll 1$ over the range span ΔR , hence $e^{-2(\alpha_m - \alpha_1)r_1} e^{-2\alpha_m(r-r_1)} \simeq 1$. Normalizing $\tilde{S}(r_1)$ to unity, one finds

$$\begin{aligned} \bar{g}(k_r, z_j) &\sim \sum_{m=1}^M \frac{\phi_m(z_s) \phi_m(z_j)}{\sqrt{k_r k_m}} \int_{r_1}^{r_2} e^{i(k_r - k_m - i\alpha_m)r} dr \\ &= \sum_{m=1}^M a_m \frac{\phi_m(z_s) \phi_m(z_j)}{k_r - k_m - i\alpha_m}, \end{aligned} \quad (10)$$

where

$$a_m = \frac{e^{i(k_r - k_m)r_2 - \alpha_m r_2} - e^{i(k_r - k_m)r_1 - \alpha_m r_1}}{ik_m}. \quad (11)$$

One finds that for a large aperture VLA, approximating $\tilde{S}(r)$ by a constant, Eq. (10) yields the same result as Eq. (4). In practice, for a finite aperture VLA, $\tilde{S}(r)$ is likely to oscillate with respect to r due to the residual mode interference. [For example, $\tilde{S}(r) \simeq s_0 + s_1 \sin k_{lm}r$ with $s_1 \ll s_0$, where $\lambda_{lm} = 2\pi/k_{lm}$ is the mode cycle distance between the l and m th modes.] In this case, it helps to smooth the oscillation by range averaging. For a non-constant $\tilde{S}(r)$, the wavenumber spectrum is given by convolving Eq. (10) with $\tilde{S}(k_r) = \int_{r_1}^{r_2} e^{-ik_r r} \tilde{S}(r) dr$, which has an effect of broadening the widths of the peaks without changing the peak position.

Note that the widths of the spectral peaks of individual modes are inversely proportional to the range span, ΔR . When the range span is equal or less than the mode cycle distance, the modes involved may not be resolved by the wavenumber analysis. In that case, one obtains an estimate of the average wavenumber of the unresolved modes. In general, this is not the case given data over a sufficient range span. The estimated mode wavenumbers (from the spectral analysis) will be denoted by $\bar{k}_m$ to differentiate them from the true wavenumbers. For the theoretical analysis to follow, we shall approximate $\tilde{S}(r)$ by a constant with $\tilde{S}(r_1)$ normalized to unity. In this case, $\bar{g}(k_m, z_j) = g(k_m, z_j)$ as mentioned in the preceding text. The spectrum value at $k_r = \bar{k}_m$ is then given by

$$g(\bar{k}_m, z_j) \sim b_m \phi_m(z_j), \quad (12)$$

where using Eq. (11), one finds²²

$$b_m = \frac{2e^{-\alpha_m r_0}}{\alpha_m k_m} \sinh\left(\frac{\alpha_m \Delta R}{2}\right) \phi_m(z_s), \quad (13)$$

where $r_0 = (r_2 + r_1)/2$ and $\Delta R = r_2 - r_1$. Note that b_m is a real number in Eq. (13). The important thing is that it does

not carry mode and range dependent phases, which would otherwise influence/bias the range estimation. From Eq. (12), one obtains an estimate of the mode depth function $\bar{\phi}_m(z_j)$ at the receiver depth z_j using the amplitude measured at the spectral peaks,

$$\begin{aligned} \bar{\phi}_m(z_j) &= \frac{g(\bar{k}_m, z_j)}{\sqrt{\sum_{j=1}^N |g(\bar{k}_m, z_j)|^2 \Delta z}} = \frac{b_m \phi_m(z_j)}{\sqrt{|b_m|^2 \sum_{j=1}^N \phi_m^2(z_j) \Delta z}} \\ &\simeq \phi_m(z_j), \end{aligned} \quad (14)$$

where the mode depth functions are normalized by $\sum_{j=1}^N |\phi_m(z_j)|^2 \Delta z = 1$ using Eq. (8) for a large aperture VLA. In practice, the estimated mode depth function $\bar{\phi}_m(z_j)$ will deviate from the true mode depth function.

To summarize, a method is proposed in this section to estimate the mode wavenumbers [using Eq. (6)] and depth functions [using Eq. (14)] for data received on a VLA from a moving source. Equation (6) can be viewed as a data-based generalized Hankel transform (at large ranges) that does not require knowledge about the source range; a large aperture VLA is required in principle. Because of the approximation involved (not knowing the source range), the estimated mode wavenumbers and mode depth functions may not be accurate. Also some of the modes may not be resolved by the wavenumber analysis. The effects of the approximation on the performance of source localization will be evaluated in Sec. III C using numerical examples, after a discussion of the data-based source localization algorithm, given in the next section, Sec. III B.

B. Data-based source localization: Narrowband signal

Given the estimated mode wavenumbers and depth functions, one can estimate the source range and depth as a function of time (by estimating the source range and depth for each snapshot or block of the data). To estimate the source location for a given (i th) block of data, one constructs a range and depth ambiguity function,

$$\text{DMMP}(r, z) = \left| \sum_{m=1}^{\bar{M}} \bar{\phi}_m(z) \frac{e^{i\bar{k}_m r}}{\sqrt{\bar{k}_m}} \sum_{j=1}^N \bar{\phi}_m(z_j) p(r_s, z_j) \Delta z \right|^2, \quad (15)$$

where $z = z_j$, $j = 1$ to N , $\bar{M}$ denotes the total number of modes (or $\bar{k}_m$) measured/resolved by the wavenumber analysis, and $p(r_s, z_j)$ is the pressure field and r_s is the unknown source range at time t_i . Equation (15) can be applied to a VLA of arbitrary configuration with the understanding that the performance will be aperture dependent. For the theoretical analyses given in the following text, we shall assume a large aperture VLA so that we can compare the data-based method with the original MMP/MFP. Using the mode orthogonality condition Eq. (8) for a large aperture VLA, Eq. (15) can be simplified to

$$\begin{aligned} \text{DMMP}(r, z) &= \frac{2\pi}{r_s} \left| \sum_{l,m=1}^{\bar{M}} \bar{\phi}_m(z) \phi_l(z_s) \sum_{j=1}^N \bar{\phi}_m(z_j) \phi_l(z_j) \right. \\ &\quad \times \Delta z \frac{e^{i[\bar{k}_m r - k_l r_s] - \alpha_l r_s}}{\sqrt{\bar{k}_m k_l}} \left. \right|^2 \\ &= \frac{2\pi}{r_s} \left| \sum_{m=1}^{\bar{M}} \phi_m(z) \phi_m(z_s) \frac{e^{ik_m(r-r_s) - \alpha_m r_s}}{k_m} \right|^2, \end{aligned} \quad (16)$$

where we assumed $\bar{k}_m = k_m$, $\bar{\phi}_m(z_j) = \phi_m(z_j)$, given a sufficient range span so that the mode wavenumbers are well resolved. Equation (15) can be interpreted as a data-based MMP, namely, $\text{DMMP}(r, z) = (r/2\pi) |\sum_{m=1}^{\bar{M}} (A_m^{\text{rplc}}(r, z))^* \times A_m^{\text{data}}|^2$, where $A_m^{\text{data}} \equiv \sum_{j=1}^N \bar{\phi}_m(z_j) p(r_s, z_j) \Delta z$ is the m th mode amplitude estimated from the data, and $A_m^{\text{rplc}}(r, z) = \sqrt{(2\pi/r\bar{k}_m)} \bar{\phi}_m(z) e^{i\bar{k}_m r}$ is the m th mode amplitude of the replica field for a hypothetical source located at (r, z) ; $z = z_j$, $j = 1$ to N . [In comparing with MMP, we assumed that given a large aperture VLA, the mode amplitudes can be obtained by multiplying the pressure field by the mode depth functions with little or no mode leakage problem. Note that no mode decomposition is used nor required for the data-based MMP as given in Eq. (15).] Now if all the modes are included, i.e., $\bar{M} = M$ by the wavenumber analysis, one finds that Eq. (16) becomes the original MMP ambiguity function. In practice, $\bar{M} < M$ as illustrated in Sec. III C. Consequently, the data-based MMP is likely to deviate from the MMP in the range-depth resolution and the sidelobe structure.

Equation (15) can also be written as

$$\text{DMFP}(r, z) = \frac{r}{2\pi} \left| \sum_{j=1}^N p_{\text{rplc}}^*(r, z_j) p(r_s, z_j) \Delta z \right|^2, \quad (17)$$

where $p_{\text{rplc}}(r, z_j) = \sqrt{2\pi/r} \sum_{m=1}^{\bar{M}} \bar{\phi}_m(z) \bar{\phi}_m(z_j) (e^{-i\bar{k}_m r} / \sqrt{\bar{k}_m})$ denotes the replica field (of a reduced number of modes) with z_j denoting the receiver depth. Equation (17) is often referred to as the Bartlett processor. Note that Eq. (17) can be applied to a short or limited aperture VLA (covering parts of the water column) as is the case for the original MFP. The difference is that for the conventional MFP, one can search for any source range and depth because $p_{\text{rplc}}(r, z_j)$ is based on a model, but for the data-based MFP, $p_{\text{rplc}}(r, z_j)$ can be calculated at any range but only at discrete depths; $\bar{\phi}_m(z)$ is only known only for $z = z_j$, $j = 1, 2, \dots, N$. In other words, the depth search is limited to the depths of the physical receivers. For application to a short VLA, one assumes that the array has enough receivers to couple to the modes excited by the source, else it may lack the ability to localize the source.

It is noted that the data-based source localization algorithm, Eq. (15), follows the same spirit as the matched field/mode processing except that no environmental acoustic information (such as SSP and/or bottom profile) is required, and no propagation model is used to calculate the replica field as a function of range and depth. It is

computationally simple and may be carried out by in-buoy processing. The drawback is that not all the modes will be resolved by the Hankel wavenumber analysis. The total number of modes estimated from the wavenumber spectral analysis is usually smaller than that allowed by the theory ($\bar{M} < M$) as will be shown by the numerical examples in the next section (Sec. III C).

C. Numerical examples

Numerical examples are given in this section to examine the properties of the data-based ambiguity surfaces, Eq. (15), for a narrowband signal to compare with that using MMP, Eq. (16). The SSP used for the simulation is similar to that collected on the New Jersey shelf in the summer time. See Fig. 1. It consists of a thermocline layer of 10 m with a sound speed of 1533 m/s. The sound speed decreases almost linearly with depth to 1478 m/s at depth of 40 m and remains at that value until a depth of 88 m. The bottom has a sound speed ~ 1650 m/s, density 1.76 g/cm³, and attenuation 0.8 dB per wavelength as used before. Two VLAs will be used in the simulation, one contains 20 elements spaced at 4 m, and the other contains 40 elements spaced at 2 m, both covering the water column from 2 to 80 m. Assume a source moves away from the VLA with a speed of 5 kn, transmitting a narrowband signal at 350 Hz. The narrowband pressure field is estimated every second, corresponding to a range increment of 2.5 m between samples of data. For the simulation, the pressure field is generated using the KRAKEN normal mode program.²³ (The source starts out at a range of 5010 m, but this information is assumed unknown.²⁴) The pressure fields, covering a range span of 490 m, are used to estimate the wavenumber spectrum for each receiver using Eqs. (6) and (7). The wavenumber spectra summed over all receivers are shown in Fig. 2(a), where the identified spectral peaks are marked by “x.” Twelve mode wavenumbers are estimated from the spectral peaks. Also shown in Fig. 2(a) is the true wavenumbers (from the normal mode program) of the lowest 16 modes, marked by symbols “+.” Compared with the true wavenumbers, one finds that the third, seventh, and 12th order modes are “absent” in Fig. 2(a), and the fifth order mode has a weak spectral value and is not identified by the peak picking program used. For the data-based MMP, which order normal mode is associated with which resolved mode is not known nor is it relevant. The number of modes measured/resolved by data is the important parameter—the larger the number of the resolved modes, the lower the sidelobe levels relative to the main lobe.

From the wavenumber spectra, the complex amplitudes associated with the spectral peaks at a given wavenumber are used to estimate the mode depth functions as a function of the receiver depth for the mode identified by the wavenumber. To determine how good is the estimation, one calculates the correlation coefficient between the estimated and true mode depth functions, defined by the maximum value of the cross correlation between the estimated and true mode depth functions, normalized by the square root of the product of the maxima autocorrelation of the estimated mode depth function and that of the true mode depth function. One finds

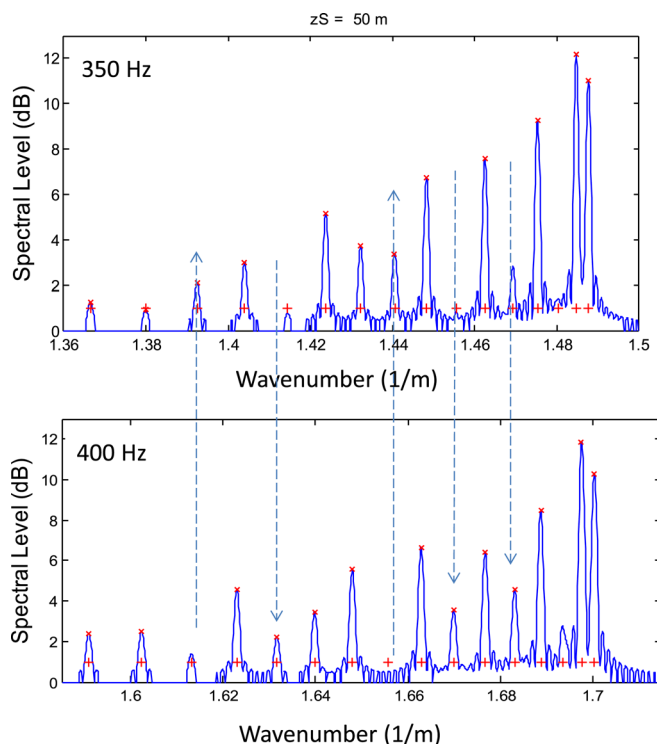


FIG. 2. (Color online) Wavenumber spectrum at 350 Hz (a) and 400 Hz (b). Spectral peaks identified by the peak picking algorithm are marked by “x” and the mode wavenumbers calculated by the KRAKEN mode program are shown by “+.” The dashed lines pointing upward indicate the modes found at 350 Hz but not at 400 Hz, and the dashed lines pointing upward show the modes found at 400 Hz but not at 350 Hz.

that the correlation coefficient is equal or greater than ~ 0.99 for the 12 resolved modes, indicating a high level of matching between the estimated and true mode depth functions. The estimated mode depth functions (not shown here) agree

well with the true mode depth functions when properly normalized.

The 12 estimated mode wavenumbers and depth functions are used to create the range and depth ambiguity surfaces using Eq. (15) for each block of data to compare with that using MMP, Eq. (16). A range and depth ambiguity surface is shown in Fig. 3, for example, for a block of data where the source is located at a range of 7000 m and depth of 50 m. Figures 3(a) and 3(b) show the MMP (or MFP) and data-based MMP ambiguity surfaces for the 40 element VLA, spaced at 2 m with a 10 dB color scale. Figures 3(c) and 3(d) show the MMP (or MFP) and data-based MMP ambiguity surfaces for the 20 element VLA, spaced at 4 m with the same color scale. (For a VLA covering the water column, MFP and MMP are basically the same.) Comparing Fig. 3(a) with Fig. 3(c) [or Fig. 3(b) with Fig. 3(d)], one finds that both give the correct source range estimation, but the depth resolution is doubled from 2 to 4 m (due to doubling of phone spacing), and the peak value is decreased by ~ 6 dB, due to the decreased signal gain (halved number of phones). Comparing the data-based MMP ambiguity surfaces [Figs. 3(b) and 3(d)] with the conventional MMP/MFP surfaces [Figs. 3(a) and 3(c)], one finds a slight increase in the sidelobe-to-main lobe ratio (i.e., increased sidelobe level relative to the main lobe level or smaller difference between the two). The reason is due to the fact that only 12 modes are used in the data-based MMP compared with 20 modes used in the conventional MMP. One notes that the range-depth ambiguity surfaces in Fig. 3 have many high level sidelobes due to the narrowband nature of the data; this is not unexpected. In general, higher sidelobe-to-main lobe ratios imply higher probability of incorrect source localization in the presence of noise and/or environmental mismatch. To improve the source localization performance and source

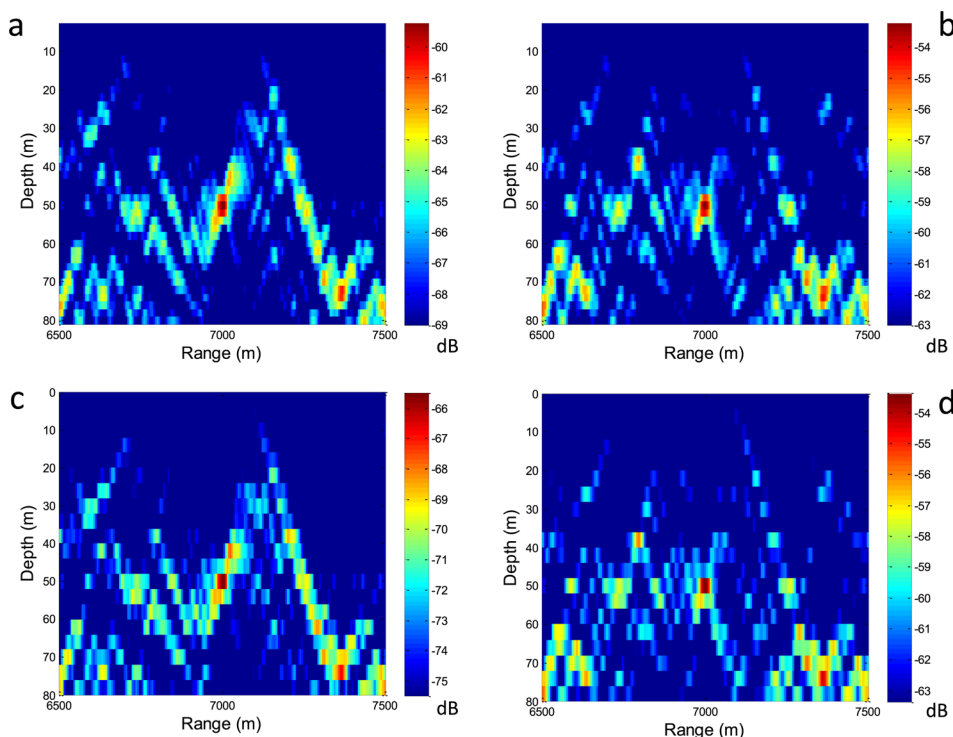


FIG. 3. (Color online) Range and depth ambiguity surface for the 40 element VLA spaced at 2 m using (a) MMP and (b) data-based MMP and for the 20 element VLA spaced at 4 m using (c) MMP and (d) data-based MMP. The input data are a 350 Hz tonal signal from a source at 7000 m range and 50 m depth.

range and depth tracking (as a function of time), conventional wisdom suggests the use of broadband signals, which are known to effectively suppress the sidelobe levels. A data-based broadband source localization approach will be presented next.

IV. BROAD BAND SOURCE LOCALIZATION

For broadband source localization, a common practice is to average the ambiguity surfaces over different frequencies within the band because sidelobe positions vary with frequencies. For multi-tonal signals emitted by a moving source, or a coherent broadband signal (with a well-defined waveform) transmitted by a moving (controlled) source, one can apply the method of Sec. III to each frequency component and add up the ambiguity surfaces. In real life, many broadband signals are (temporally) incoherent like the broadband noise radiated by a ship, for which the preceding Hankel transform method cannot be applied because the noise-like signal contains random (time-dependent) phases for each frequency component. Nonetheless, each snapshot of the noise-like data can still be used to provide an estimate of the source location because even though the noise-like signal waveform is random and unknown, the signals received on the VLA are spatially correlated between the receivers. The ambiguity functions so obtained can then be averaged over many snapshots to minimize the variance of the estimation. For data-based approaches to broadband source localization, one needs to know the mode wavenumbers and depth functions for the broadband frequency components. Because they could not be directly estimated from the broadband noise data, we propose to estimate them by interpolation from the mode wavenumbers and depth functions of the tonal frequencies, assuming that the source also radiates narrowband tones. The interpolation of the mode wavenumbers is derived based on the mode equation for the mode group velocity,²⁵ and the interpolation of the mode depth functions is based on the WKB approximation.^{25,26}

A. Wavenumber estimation across frequencies

To estimate the mode wavenumbers over broad band frequencies, we use the following equation for the mode group velocity,²⁵

$$\frac{1}{U_m} \equiv (2\pi)^{-1} \frac{\partial k_m}{\partial f} = 4\pi^2 \frac{f}{k_m} \int_0^H \frac{\phi_m^2(z)}{\rho(z)C^2(z)} dz, \quad (18)$$

from which one obtains

$$k_m^2(f) - k_m^2(f_0) = \gamma_m(f^2 - f_0^2), \quad (19)$$

where $\gamma_m = (2\pi)^3 \int_0^H [\phi_m(z)^2 / \rho(z)C^2(z)] dz$ is a mode-dependent function for a given SSP and can be determined from the data given wavenumber measurements for two tonal frequencies, $\gamma_m = [k_m^2(f_1) - k_m^2(f_0)](f_1^2 - f_0^2)^{-1}$. One then obtains a data-based interpolation/extrapolation formula,

$$k_m^2(f) - k_m^2(f_0) = [k_m^2(f_1) - k_m^2(f_0)] \frac{f^2 - f_0^2}{f_1^2 - f_0^2}, \quad (20)$$

where we dropped the overhead bar for the estimated mode wavenumbers.

To check the accuracy of Eq. (20), we create simulated data at 400 Hz generated by a moving source using the same environment and geometry as discussed in Sec. III C. We then estimate the mode wavenumbers at 400 Hz following the same approach for 350 Hz (the data-based Hankel transform) as described in Sec. III C. The wavenumber spectrum is shown in Fig. 2(b) to compare with that at 350 Hz shown in Fig. 2(a). By changing the wavenumber scale using Eq. (20), one can associate the (mode) orders between the two sets of wavenumbers between Figs. 2(a) and 2(b) as shown in Fig. 2. One finds at 400 Hz that 13 modes, marked by “x,” are resolved by wavenumber analysis out of 16 true modes marked by “+.” (For completeness, we shall state the mode order for each resolved mode based on that determined from the normal mode program. At 400 Hz, we find that the third, ninth, and 14th order modes were weakly excited by the source.) For the data-based analysis, the true mode orders are not known nor are they important. The task is to associate the resolved modes at 350 Hz with those at 400 Hz. For that purpose, we shall name the mode number for the resolved modes sequentially in terms of decreasing wavenumbers. Comparing Fig. 2(a) with Fig. 2(b), one finds that for the number 6 and 10 modes at 350 Hz, there are no counter parts at 400 Hz. See the dashed line with arrows pointing upward in Fig. 2. (They are physically the ninth and 14th order modes). Similarly, for the number 4, 6, and 10 modes at 400 Hz, there are no counter parts at 350 Hz. See the dashed line with arrows pointing downward in Fig. 2. (They are physically the fifth, seventh, and 12th modes). For the remaining modes, one can associate the following numbered modes [1,2,3,4,5,7,8,9,11,12] at 350 Hz with the following numbered modes [1,2,3,5,7,8,9,11,12,13] at 400 Hz. In other words, only ten pairs of the modes are common to both 350 and 400 Hz frequencies and can be used for interpolation for frequencies in between 350 and 400 Hz (of the physically 16 pairs of modes). The price paid for data-based method is a reduced number of modes available for source localization.

The mode wavenumbers for frequencies between 350 and 400 Hz can be estimated from the wavenumbers measured at 350 and 400 Hz using Eq. (20) (for those modes that are common to both). The results are shown in Fig. 4, marked by “+” for nine frequencies from 355 to 395 Hz to compare with the mode wavenumbers calculated by the normal mode program, marked by “x” for each frequency. One finds good agreement between numerically estimated wavenumbers and true wavenumbers.

B. Depth function estimation across frequencies

To estimate the mode depth function over broad frequencies, we assume the WKB approximation.^{25,26} The WKB modes between the ray turning points in a sound channel are given by

$$\phi_m(z) = A_m(z) \sin \eta_m(z), \quad (21)$$

where $A_m(z)$ is a slowly varying amplitude given by $A_m(z) = k_{v,m}^{-1/2}(z) = [\omega^2/C^2(z) - k_m^2]^{-1/4}$. Here $k_{v,m}(z)$ is the

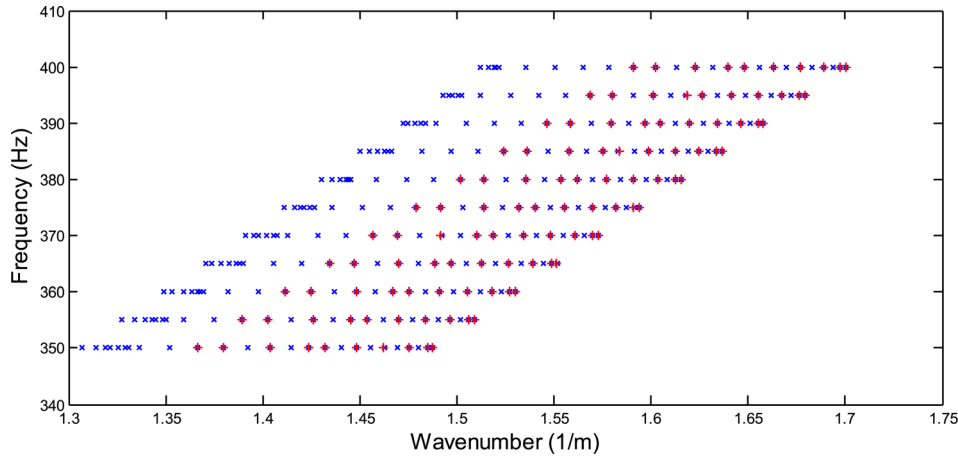


FIG. 4. (Color online) Mode wavenumbers at nine frequencies from 355 to 395 Hz are estimated from the mode wavenumbers at 350 and 400 Hz using Eq. (20) and marked by “+” to compare with that calculated by the KRAKEN program, marked by “x”.

mode vertical wavenumber, and the WKB phase is given by $\eta_m(z) = \eta_a + \int_a^z k_{v,m}(z') dz'$, where the quantity η_a is the phase at the upper turning point. The WKB modes can be interpreted as the interference pattern created by two plane waves. One travels upward at a grazing angle $\theta(z)$, given by $k_{v,m}(z) = k(z) \sin \theta(z)$, where $k(z) = \omega/C(z)$, and the other travels downward at the same angle. To maintain the same number of interference nulls (i.e., the mode number) as frequency is changed, the angle of propagation must change accordingly. Note that the mode wavenumber $k_m(z) = k(z) \cos \theta(z)$ generally increases with frequency, and hence $k_{v,m}(z)$ generally decreases with frequency. Equation (21) suggests that the mode depth functions at different frequencies maintain the same number of nulls or axis-crossing points where $\phi_m(z) = 0$, but the spatial extent (coverage) decreases with increasing frequencies due to decreasing $k_{v,m}(z)$, as observed based on numerical calculations of mode depth functions.²⁵ Generally speaking, the mode depth functions change slowly within a narrow frequency band between f_0 and f_1 , if $(f_1 - f_0) \ll (f_1 + f_0)$. In this case, to estimate the mode depth functions at frequencies between f_0 and f_1 , we propose the following procedures: (1) Determine whether the mode depth functions estimated at the two frequencies f_0 and f_1 have the same order (number of intensity nulls or maxima) by cross-correlating the mode depth functions. The pairs of modes having the largest normalized cross-correlation coefficients are hypothesized to have the same order. (2) Determine the mode depth functions for the ordered pairs at frequencies between f_0 and f_1 by linear interpretation.

For the mode depth functions estimated at 350 and 400 Hz, we plot the absolute values of the mode depth functions for the [1,2,3,4,5,7,8,9,11,12] modes at 350 Hz and compare with the mode depth functions for the [1,2,3,5,7,8,9,11,12,13] modes at 400 Hz one by one. We find that for each pair of modes (between the 350 and 400 Hz), the amplitudes have the same number of peaks and valleys as expected for the same order of mode, confirming that the modes paired based on the wavenumber scaling rule (Sec. IV B) were correctly associated. Also the cross correlation of the mode depth functions between each pair of associated modes shows higher correlation than cross correlation with other non-associated modes, confirming that the mode

association is correct. It is noted that when the first (order) mode is not found in the wavenumber spectrum (as for a surface source), the mode association by wavenumber scaling may be ambiguous because in this case, one does not have the lowest order mode to serve as a reference. In that case, cross correlation of the estimated mode amplitudes can be used to associate the modes for those mode depth functions that are adequately (Nyquist) sampled.

The mode depth functions at frequencies between 350 and 400 Hz are obtained by linearly interpolating the mode depths functions estimated at 350 and 400 Hz for the ten pairs of modes. The interpolated mode depth functions are in good agreement with that obtained from the KRAKEN normal mode program. There are minor differences but they do not affect source localization. The results are not shown here.

For two frequencies f_0 and f_1 that are widely separated, one can estimate the WKB mode depth function $A_m(z) \sin \eta_m(z)$ by evaluating the vertical mode wavenumber $k_{v,m}(z, f)$ at frequency f . One finds, using Eq. (19),

$$\begin{aligned} k_{v,m}^2(z, f) - k_{v,m}^2(z, f_0) &= \left[\frac{4\pi^2}{C^2(z)} - \gamma_m \right] (f^2 - f_0^2) \\ &= [k_{v,m}^2(z, f_1) - k_{v,m}^2(z, f_0)] \frac{f^2 - f_0^2}{f_1^2 - f_0^2}, \end{aligned} \quad (22)$$

where $k_{v,m}(z, f_0)$ and $k_{v,m}(z, f_1)$ are estimated from the mode depth functions at $z = z_i$, $i = 1, 2, 3, \dots, N$. Interpolation of mode depth functions in this case is more complicated.

C. Broadband ambiguity surfaces

One obtains the broadband ambiguity surfaces by summing the ambiguity surfaces over the narrowband frequencies. For the numerical simulations, we choose 11 frequencies between 351 and 399 Hz. Repeating the narrowband source localization for 11 frequencies (for the source at 7000 m range and 50 m depth, see Sec. III C and Fig. 3), the broadband ambiguity surface, summing over the 11 ambiguity surfaces, are shown in Figs. 5(a) and 5(b) for the 40 element VLA, spaced at 2 m, using the conventional

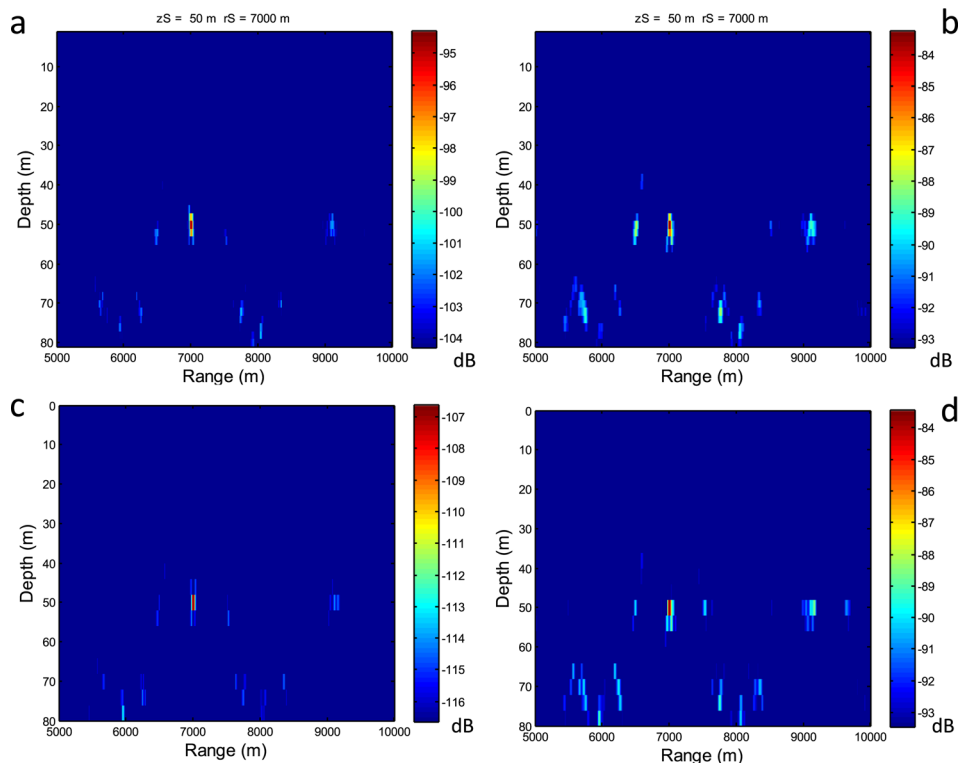


FIG. 5. (Color online) Broadband range and depth ambiguity surface, obtained by summing the narrowband ambiguity surfaces for 11 frequencies between 350 and 400 Hz, are shown for the 40 element VLA spaced at 2 m using (a) MMP and (b) data-based MMP and for the 20 element VLA spaced at 4 m using (c) MMP and (d) data-based MMP. The source is located at 7000 m range and 50 m depth.

MMP/MFP and data-based MMP methods, respectively. The color scale shows a 10 dB dynamic range for both of them. One observes that the source range and depth are clearly identified using a broadband signal. Comparing Fig. 5(b) with Fig. 5(a), the sidelobe-to-main lobe ratios are a little bit higher (~ 1 dB) for the data-based MMP than the conventional MMP. One of the reasons is that the data-based MMP uses only ten modes, whereas the conventional MMP uses 20 modes. For this simulation, the ratio of the input signal level averaged over receivers and frequencies with respect to the noise level averaged the same way is about 20 dB. The ambiguity surfaces shown in Fig. 5 are for one snapshot of broadband data. In this case, there is interference between the signal and noise; this contributes to higher sidelobe levels. It can be verified that the ambiguity surface averaged over many snapshots has a reduced sidelobe level; the SNR at the main peak compared with the input SNR is the array gain. The detection problem is outside the scope of this paper.

Figures 5(c) and 5(d) show the broadband MMP/MFP and data-based MMP ambiguity surfaces for the 40 element VLA, spaced at 4 m. One finds that the 20 element VLA provides equally good source location in terms of sidelobe structure, albeit with a poorer depth resolution (from 2 to 4 m) and smaller signal gain. The sidelobe-to-main lobe ratios are approximately the same comparing Figs. 5(c) to 5(a) and Figs. 5(d) to 5(b).

Figures 6(a) and 6(b) show the broadband MMP/MFP and data-based MMP ambiguity surfaces for a near surface source located at 4 m depth using the 40 element VLA, spaced at 2 m. The same 11 frequencies from 355 to 395 Hz were used. For the data-based MMP, the mode wave-number and depth functions are estimated for 11 modes

excited by the moving surface source in the same manner as discussed in Sec. III. (The 4 m source excites predominantly the seventh to 16th modes with intensities peaked at the 11th and 12th modes.) Figures 6(c) and 6(d) show the broadband MMP/MFP and data-based MMP ambiguity surfaces for the 20 element VLA, spaced at 4 m, for the same 11 frequencies. One notes that the ambiguity surfaces for the shallow source (Fig. 6) have higher sidelobes compared with that for the 50 m source (Fig. 5). The reason is that the high order modes generated by the near surface source are not so well resolved by the VLA.

V. SHORT VLA

A VLA, which spans only a portion of the water column, is desirable from the system point of view (reduced cost, easy deployment, and extended life time when powered by battery) if it can be used to estimate the source location reliably for the targets of interest. The conventional approach by averaging the ambiguity surfaces over frequencies requires that the main lobe locations are consistent between frequencies. Now at a fixed frequency, the range-depth ambiguity surface for a short/sparse VLA generally contains more high level sidelobes than that for a VLA covering the full water column because the mode functions are (severely) under sampled. This is likely to lead to source localization errors in the presence of environmental mismatch (in real data). If the mismatch problem goes away using the data-based source localization method, the utility of a short VLA may become attractive again, for example, when the VLAs are deployed as part of a distributed system.

As an example, we show in Fig. 7 the broadband range and depth ambiguity surfaces obtained by averaging over

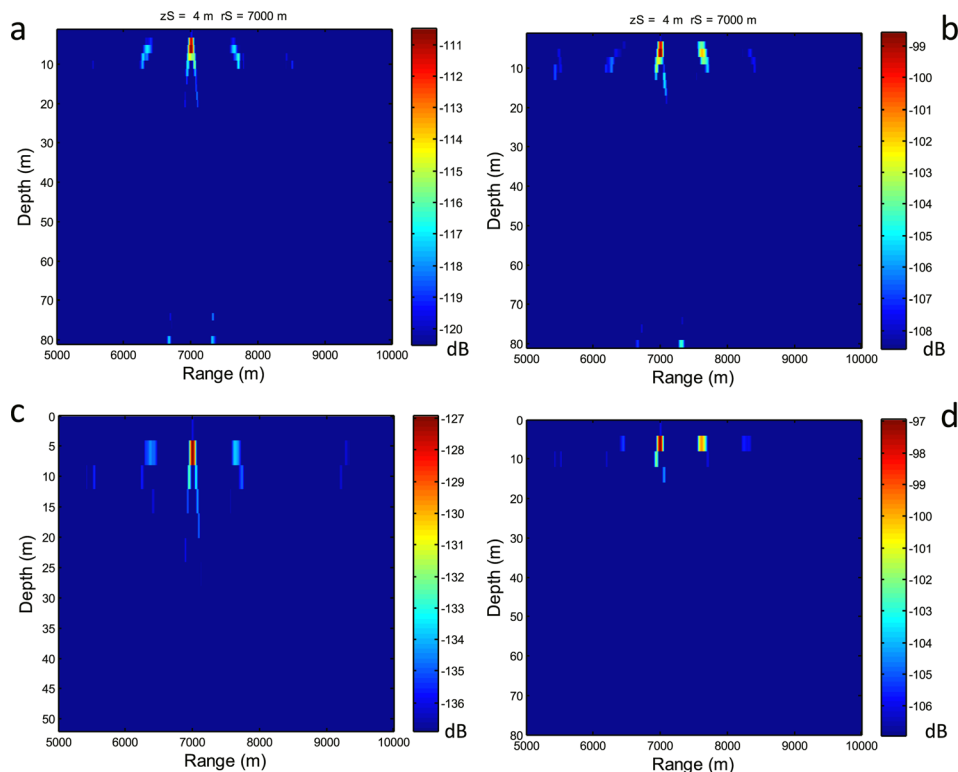


FIG. 6. (Color online) Broadband range and depth ambiguity surfaces, obtained by summing the narrowband ambiguity surfaces for 11 frequencies between 350 and 400 Hz, are shown for the 40 element VLA spaced at 2 m using (a) MMP and (b) data-based MMP and for the 20 element VLA spaced at 4 m using (c) MMP and (d) data-based MMP. The source is located at 7000 m range and 4 m depth.

11 frequencies (as discussed in Sec. IV C) for a VLA with 13 elements extending from 30 to 78 m at 4 m spacing. Figure 7(a) is for the conventional MFP, using 20 modes for a given frequency, and Fig. 7(b) is obtained by the data-based MFP [Eq. (17)], using ten modes for each frequency. One finds that the broad band MFP ambiguity surface has higher sidelobe-to-main lobe ratios for the short VLA [Fig. 7(a)] compared with that using that using the full water column array [Fig. 5(d)]. On the other hand, the ambiguity surfaces for the short VLA have similar sidelobe-to-main lobe ratios for the data-based MFP as well as the conventional MFP.

While the data-based MFP can, strictly speaking, only be used to search for sources the depth of which is within the depth span of the VLA, one may extend the depth search for a short VLA by extrapolating the mode depth functions using the WKB modes, Eq. (21), determined using the mode depth functions measured at the

receiver depths. This topic will not be pursued in this paper.

VI. SUMMARY AND DISCUSSIONS

In this paper, we developed the data-based matched mode/field source localization algorithms for a moving source, using mode wavenumbers and depth functions estimated directly from the data. The method is in principle free of the environmental mismatch problem because the mode replicas are estimated from the same data used to localize the source, and the approximations made (in deriving the data-based algorithms) are very general and independent of the environments. The proposed method does have some inherent disadvantages: (1) It uses a smaller number of modes than theoretically possible because some modes are not resolved in the measurements and (2) the depth search is limited to the depth covered by the receivers. The first

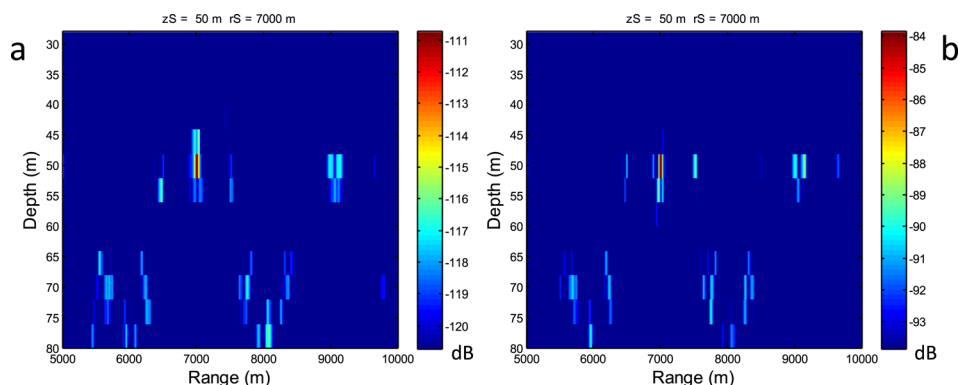


FIG. 7. (Color online) Broadband range and depth ambiguity surfaces, obtained by summing the narrowband ambiguity surfaces for 11 frequencies between 350 and 400 Hz, are shown for the 13 element VLA, extending from 30 to 78 m, using (a) MFP and (b) data-based MFP. The source is located at 7000 m range and 50 m depth.

order issue is whether and how the approximations and the limitations affect the source localization performance. To address this issue, we conduct numerical simulations to study/evaluate the performance difference between the data-based and the original MMP/MFP. One finds that for the cases studied, the sidelobe-to-main lobe ratio increases by ~ 1 dB comparing the data-based MMP range/depth ambiguity surfaces with that of the original MMP/MFP. The results suggest that the data-based MMP/MFP is capable of localizing the source with a performance comparable to that of the original MMP/MFP. The advantage is that the data-based MMP/MFP does not require any environmental information nor does it require a propagation model to calculate the replica fields.

Specifically, for the narrowband signals, we proposed a data-based generalized Hankel transform to estimate the mode wavenumbers and mode depth functions. The original Hankel transform, used previously to estimate the mode wavenumbers, requires knowledge of the source range. To compensate for the cylindrical spreading loss of the signal (without knowing the source range), we used the depth-averaged signal intensity measured from data. This algorithm requires in principle a large aperture VLA. It can be applied to short VLAs by range-averaging the depth-averaged intensities. From the Hankel-transformed wavenumber spectra, one estimates the mode wavenumbers from the spectral peaks. In addition, the mode depth functions can be estimated based on the complex amplitudes of the spectral peaks for each receiver. For the numerical simulation, one assumes that the source travels 450 m radially within 3 min observation time (at a speed 5 kn). One finds that 12 modes (of 16) can be resolved by the generalized Hankel transform. The data-based range-depth ambiguity surface shows a slightly higher sidelobe-to-main lobe ratio due to the smaller number of modes used and (spectral) estimation errors.

To localize a target using the noise-like signals radiated by the target, we assume that the target also radiates narrowband tonal signals as is often the case. Given the wavenumbers estimated at the tonal frequencies, we use an equation for the mode group velocity to interpolate the mode wavenumbers for the broadband frequency components. To estimate the mode depth functions for the broadband signals, we assume the WKB approximation. Numerical examples are given for a broadband signal by summing incoherently the range-depth ambiguity surfaces over 11 frequencies between 350 and 400 Hz. Only ten modes are used out of 16 normal modes because only ten modes are resolved at both the 350 and 400 Hz. This reduced number of modes is responsible for a ~ 1 dB higher sidelobe-to-main lobe ratio for the data-based approach compared the original MMP/MFP.

The proposed method has a potential to be applied to real data and be free of the environmental mismatch problem, noting that certain aspects of the (generalized) Hankel transform have already been tested against data. The state of the art and the processing issues are summarized in the following text.

- (1) First, one notes that the original Hankel transform has been successfully applied to experimental data in

different waters including weakly range-dependent environments where adiabatic propagation can be assumed. In testing the original Hankel transform against data, the source range was assumed known.^{19–21} For the algorithm proposed here, the source range is assumed unknown, but the range *span* is assumed to be estimated from data for a source traveling a straight line with a constant speed. [For a non-radial run, the range span is given by $\Delta R = \int_0^T v(t)dt = \bar{v}T$, where the radial velocity v (measured from the Doppler shift) is changing with time.²⁷] It is likely that the range span ΔR may only be approximate due to uncertainty in the Doppler estimation. The question is how this will affect the source range estimation.

Assuming that the velocity estimate is off by a factor α , so will the range span be. Based on Eq. (6), one finds that if the estimated range span is in error by a factor of α , the wavenumber is reduced/enlarged by a factor $1/\alpha$, which in turn implies that the search range is scaled up/down by a factor of α . This implies that the range estimation for the source will have an error on the order of $(1 - \alpha)$ times the true range. It is noted that the scale change is not expected to affect the distribution or the “landscape” of the ambiguity surface. In other words, one expects that the source will be consistently localized over time despite the scale change (or shift in target position) because there is no environmental mismatch. The fact that the peaks of the ambiguity surfaces will form a (target) track as a function of time (with a small variance) is very important from the system point of view; it provides an important cue on whether a target exists or not. (The problem with the conventional MFP is that the source localization results are often inconsistent over time in the presence of environmental mismatch, making the results questionable.)

- (2) For real data, the radiated tonal signals often encounter a time-varying phase change due to the unsteady source motion; the tonal signal is no longer a pure sinusoidal. For example, a source towed behind a ship is affected by the motion of the towing ship, which moves up and down following the waves, inducing a jittering motion (acceleration/deceleration) on the tow body. Before applying the (generalized) Hankel transform, the fluctuating phase due to source acceleration/deceleration needs to be removed. Likewise, for a source moving in a non-radial direction, the radial velocity is no longer a constant. In this case, although the phase introduced by the source acceleration deceleration is deterministic, it can induce error if the data is processed (integrated) over an extended period of time. This problem was recently addressed by Walker *et al.*,²⁷ who proposed a high resolution nonlinear frequency tracking technique to measure the instantaneous frequency response of the signal. Based on that, one introduces a transform to remove unwanted phase (distortion). This method was (successfully) applied to tonal data from the SWellEx 96 experiment to estimate the Doppler shifts and mode depth functions of individual modes.²⁷ From the signal processing point of view, what is required here is the equivalent of a phase locked loop²⁸ (used in radio and acoustic

communications) to remove the random phase fluctuation and track the (slowly varying) range dependent phase. The phase compensation mechanism is discussed in the appendix.

- (3) For real data, one may not be able to integrate the data over a large range span due to signal fluctuations. Given a limited range span, certain modes may not be resolved by the (generalized) Hankel transform. This is likely to introduce performance degradation (increased sidelobe-to-main lobe ratio) but is not likely to introduce mismatch because one is matching the unresolved (composite) mode deduced from the data against the data. (Consequently, the robustness of the result may be maintained.) For short range-span data, one can apply high resolution methods as proposed in the literature.²⁹ In that respect, one notes that wavenumber estimation is no different from spectral estimation. Various techniques proposed/developed in the literature for spectral estimation of real world data (such as maximum entropy, maximum likelihood estimation methods) can be readily applied for wavenumber estimation.

The preceding discussions indicate that while various aspects of the wavenumber estimation method have been investigated before, the use of the generalized Hankel transform still remains to be demonstrated with real data. It is noted that so long as the mode wavenumbers for the energetic modes can be resolved and estimated from the data, the corresponding mode depth functions can be estimated accordingly without much loss of information. Given the estimation of the mode wavenumbers and depth functions from data, the proposed method has a potential to estimate the source range and depth from data and be free of the environmental mismatch problem. Numerical simulations conducted in this paper suggest small performance degradation in source localization between the data-based MMP/MFP and the original MMP/MFP (without mismatch). The actual performance degradation is likely to be not only dependent on the array aperture but also on the acoustic environment. It also will depend on the accuracy of the signal processing approaches in estimating the mode wavenumbers.

ACKNOWLEDGMENTS

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APPENDIX: PHASE ESTIMATION AND COMPENSATION

It is well known that source accelerations (decelerations) lead to Doppler broadening of the signal frequency. The signal from a moving narrowband source with an angular frequency ω_c can be expressed as

$$p(r, z; t) = \text{Re} \left[\sum_{m=1}^M \sqrt{\frac{2\pi}{k_m r}} \phi_m(z_s) \times e^{i(\omega_s t - ik_m r(t) - \alpha_m r(t) - i\pi/4)} \phi_m(z) \right], \quad (\text{A1})$$

where ω_s is the transmitted frequency, r is the source range, $r(t) = r_k + vt + 1/2at^2 + \dots = r_k + vt + \tilde{r}(t)$, where r_k is the initial range associated with the k th block of data, $\bar{v}$ is the average radial velocity, and $\tilde{r}(t)$ contains the source acceleration a . The spectrum amplitude obtained by a Fourier transform of the k th block of data evaluated at the Doppler shifted frequency $\omega_c = \omega_s - k_0 \bar{v}$, where $k_0 = \omega_s/C_0$, is given by

$$p(r_k, z) = \sum_{m=1}^M \sqrt{\frac{2\pi}{k_m r_k}} \phi_m(z_s) e^{-ik_m r_k - \alpha_m r_k - i\pi/4} \times \phi_m(z) e^{-ik_m \tilde{r}(t)} + O(v(k_0 - k_m)). \quad (\text{A2})$$

Compared with Eq. (3), the extra term $e^{-ik_m \tilde{r}(t)}$ contains the phase distortion due to the source acceleration or jittering. To compensate for this phase distortion, one needs to estimate the instantaneous frequency of the signal. To this end, Walker *et al.* proposed a time domain least mean square fit between the received short-time duration signal and a sinusoidal signal. Dividing the block of data into many short time durations, the instantaneous frequency $\hat{\omega}$ at time t_l is obtained by minimizing the least square error summed/averaged over the receivers,

$$Q(\hat{\omega}) = \sum_{j=1}^N \int_{t_l}^{t_l+\Delta} |P(r, z_j; t') - D_j \sin(\hat{\omega} t' + \sigma_j)|^2 dt', \quad (\text{A3})$$

where $P(r, z_j; t')$ is the narrowband time series for the j th receiver obtained by applying a narrowband filter to the received data, and D_j and σ_j are unknown parameters to fit the data, which are time dependent. Having obtained the instantaneous frequency $\hat{\omega}(t) \simeq \omega_s - k_0 \bar{v} - k_0 \dot{\tilde{r}}(t)$, one integrates $\dot{\tilde{r}}(t) = [\omega_c - \hat{\omega}(t)/k_0] - \bar{v}$ over time to estimate the phase distortion $\tilde{r}(t) = \int_0^t \dot{\tilde{r}}(t') dt'$. One then multiplies the data of Eq. (A2) by $e^{ik_m \tilde{r}(t)}$ to remove the phase distortion.

An alternative approach is to employ a phase locked loop²⁸ that estimates the phase compensation $e^{ik_m \tilde{r}(t)} \equiv e^{i\theta(t)}$ by minimizing the mean square error

$$MSE = \sum_{j=1}^N \int_{t_l}^{t_l+\Delta} |P(r, z_j; t') - D_j \sin(\omega_c t' + \theta(t))|^2 dt', \quad (\text{A4})$$

where the desired signal is locked to the Doppler shifted frequency ω_c .

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